

Dinupa Devinda

AI/ML-Focused Engineering Graduate

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PROFILE

AI/ML-focused engineering graduate with an interdisciplinary background in electrical and electronic engineering, computer science, and mathematics. Experienced in developing and evaluating computer vision, audio classification, sensor-based machine learning, language model, and retrieval-augmented generation systems. Built ML workflows spanning data preparation, model training, evaluation, API serving, cloud experimentation, and edge deployment. Interested in building reliable AI/ML systems for real-world engineering and technology applications.

EDUCATION

Sri Lanka Institute of Information Technology (SLIIT)

BSc. Engineering (Hons) in Electrical and Electronic Engineering

Second Class Upper Division

2022 – 2026 | Sri Lanka

University of Kelaniya

BSc. in Physical Science

(Subjects: Computer Science, Pure Mathematics, Applied Mathematics)

2023 – 2026 | Sri Lanka

SKILLS & EXPERTISE

Programming Languages: Python, C/C++, Java, JavaScript, TypeScript, C#, SQL, PIC Assembly

Machine Learning & AI: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, Optuna, NumPy, Pandas, joblib, OpenCV, YOLO, ONNX Runtime, TensorFlow Lite, NCNN, Chroma, Cloudflare Vectorize

Backend, Cloud & Data: FastAPI, Docker, AWS EC2, Amazon S3, Amazon SageMaker, Cloudflare Workers, Next.js, React, NestJS, Spring Boot, Supabase, PostgreSQL/PostGIS, MongoDB, SQL Server

Development, DevOps & IoT: Git, GitHub Actions, MQTT/MQTTS, ThingsBoard

Technical Areas: Machine Learning, Deep Learning, Transformers, Language Modeling, LLM Applications, Retrieval-Augmented Generation (RAG), Embeddings, Vector Search, Model Serving, Computer Vision, Object Detection, Audio Classification, Time-Series Classification, Sensor Fusion, Edge AI, Feature Engineering, Model Evaluation, Data Analysis, Full-Stack Development, REST API Development, Cloud Deployment

PROFESSIONAL EXPERIENCE

Research and Development Engineering Intern

Sri Lanka Telecom – Digital Lab | Hybrid | June 2025 – September 2025

- Developed C/C++ firmware libraries for ESP32-based edge devices with A7680C 4G modem AT-command handling, MQTT/MQTTS, HTTP, SMTP, SMS, and call support.
- Built device-to-cloud telemetry pipelines using MQTT/HTTP and structured JSON payloads for live monitoring, alerts, remote status reporting, and downstream analysis.
- Configured ThingsBoard and Grafana dashboards and custom web portals; used MQTT Explorer and HiveMQ to test MQTT topics, validate JSON payloads, and debug telemetry, while implementing GitHub Actions and CI/CD workflows.

Research and Development Inplant Trainee – Electronic Manufacturing Services Floor

Variosystems (Pvt) Ltd | Onsite | June 2024 – September 2024

- Built C/C++ based automation and embedded-control projects for a manufacturing environment, including an Automated Guided Vehicle, flame-treatment automation upgrade, and automated PCB conveyor system.
- Gained hands-on exposure to production data flow, SMT/THT manufacturing, ICT and functional testing, SPEA 3030/4050 test systems, FactoryLogix workflows, and NPI processes.

PROJECTS

Vehicular Black Box: Vehicle Violation Detector

Machine Learning, Edge AI, Sensor Fusion | Nov 2025 – May 2026

- Built and integrated a Raspberry Pi-based vehicle violation detection system combining computer vision, audio classification, IMU-based ML, GPS/GSM communication, backend APIs, dashboard visualization, and driver scoring, with documented datasets, model cards, evaluation summaries, and a BYD Atto BMI160 IMU dataset published on [Zenodo/Kaggle](#).

Road Sign Detection: Developed a two-stage computer vision inference pipeline using YOLO, NCNN, ONNX Runtime, and OpenCV for Raspberry Pi deployment, achieving 0.783 precision and 0.596 F1 across 9,062 test images.

Horn Detection: Built an audio classification model using log-mel spectrogram feature extraction, TensorFlow/Keras CNN training, and TensorFlow Lite deployment for live microphone inference.

Crash Detection: Developed crash-detection models using PyTorch CNN audio inference and TensorFlow/Keras CNN-GRU IMU time-series inference for multi-signal event detection, achieving F1 scores of 0.828 and 0.947, respectively.

Lane Change Detection: Built an IMU time-series classification model using accelerometer/gyroscope windowing, feature engineering, XGBoost training, and joblib deployment artifacts, achieving 0.981 recall and 0.668 F1 in internal validation.

Aggressive Driving Detection: Developed an Optuna-tuned XGBoost classifier using yaw-rate and acceleration windows, achieving 0.790 macro F1 and 0.862 ROC AUC in final validation.

GPT-Style Language Model

PyTorch, Transformer Decoder, FastAPI, Docker, AWS EC2, Amazon S3, SageMaker | Aug 2026

- Built and deployed a 63.49M-parameter character-level GPT-style model in PyTorch with 12 decoder layers and 8 attention heads, implementing token and positional embeddings, masked self-attention, feed-forward layers, LayerNorm, residual connections, and configurable autoregressive sampling.
- Deployed a Dockerized FastAPI inference API on AWS EC2 for live CPU inference, stored the checkpoint in Amazon S3, validated it through a SageMaker notebook workflow, and integrated the API with the portfolio through a Cloudflare Worker proxy.

AI Portfolio Assistant – Hybrid RAG Chatbot

Next.js, Cloudflare Workers, Docker, Gemini API, Cloudflare Vectorize | Jul 2026

- Built a hybrid RAG portfolio assistant using Gemini, Cloudflare Workers, and Vectorize, combining semantic and keyword retrieval with reciprocal-rank fusion, source validation, and security controls; evaluated retrieval on a 20-query offline test set and Dockerized a local Python/Chroma environment for reproducible indexing, testing, and evaluation.

ServiceLink: On-Demand Local Hiring Platform

Full-Stack Development, REST APIs, Cloud Deployment | Oct 2025 – Jan 2026

- Built the original academic service marketplace prototype using Java 17, Spring Boot REST APIs, Spring Security/JWT authentication, MongoDB, React, Vite, React Router, React Query, and Axios, implementing role-based access, service listings, bookings, admin workflows, and API-driven frontend/backend communication.
- Extended the same marketplace concept for implementation using a TypeScript monorepo, Next.js frontend, NestJS REST API, Supabase authentication, PostgreSQL/PostGIS, private file storage, GitHub Actions CI/CD, Cloudflare Workers frontend deployment, and Dockerized Google Cloud Run backend deployment.

Other Projects: Built IoT, embedded, automation, backend, and academic software projects including an ESP32 A7680C 4G communication/ThingsBoard monitoring system, Raspberry Pi Smart Fish Habitat Management System, Arduino Automated Guided Vehicle, automated PCB conveyor system, DS1302 RTC assembly project with PIC16F877A/MAX6952/Proteus, BRMS Spring Boot API, Java Music Streaming Management System, C file-based Public Transport database, C#/.NET Music Player App, and PHP/MySQL EduPress Store.

CERTIFICATIONS

- PyTorch Bootcamp (OpenCV University, 2025)
- AWS Essential Training for Developers (LinkedIn Learning, 2026)
- AI/ML Engineer Stage 2 and Stage 1 (SLIIT, 2025 and 2023)
- Machine Learning with Python Professional Certificate (Anaconda, 2026)
- Hands-On AI: Introduction to Retrieval-Augmented Generation (LinkedIn Learning, 2026)
- Artificial Intelligence Foundations: Machine Learning (LinkedIn Learning, 2026)
- Complete Guide to Generative AI for Data Analysis and Data Science (LinkedIn Learning, 2026)
- Machine Learning: Foundations, Decision Trees, Logistic Regression, k-Means Clustering (LinkedIn Learning, 2026)
- Processing Text with Python Essential Training – NASBA CPE (LinkedIn Learning and NASBA, 2026)
- Certificate in Business Accounting (CIMA, 2023)